SIMATS ENGINEERING

# SIMATS Online Programming Lab

C PROGRAMMING

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## CO3 - AT1- Analytcial Problem solving

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QUESTIONS

Q1
CO3 - AT1 - Analytical Problem Solving
20 marks

Q2
CO3 - AT1 - Analytical Problem Solving
20 marks

Q3
CO3 - AT1 - Analytical Problem solving
20 marks

Q4
CO3 - AT1 - Analytical Problem Solving
20 marks

Q5
CO3 - AT1 - Analytical Problem Solving
20 marks

Q1 **CO3 - AT1 - Analytical Problem Solving** 20 marks

1. Given an N × N matrix, Write a C Program to determine whether it is:

a) Symmetric

b) Upper triangular

c) Lower triangular

d) Diagonal matrix

e) Identity matrix

f) Display all applicable classifications.

Your Code

```c
#include <stdio.h>
#include <stdbool.h>

int main() {
    int n, i, j;
    bool isSymmetric = true;
    bool isUpperTriangular = true;
    bool isLowerTriangular = true;
    bool isDiagonal = true;
    bool isIdentity = true;
```

Input

stdin input

Output

## CO3 - AT2 - Class Test

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QUESTIONS

Q1
CO3 - AT2 - Class Test
20 marks

Q2
CO3 - AT2 - Class Test
20 marks

Q3
CO3 - AT2 - Class Test
20 marks

Q4
CO3 - AT2 - Class Test
20 marks

Q5
CO3 - AT2 - Class Test
20 marks

5/5 answered

Q1 **CO3 - AT2 - Class Test** 20 marks

1. Write a program to count the frequency of each element in an array.

Sample Input: 2 2 1 5 1 3 3 (7 elements)

Sample Output: 1: 2 times, 2: 2 times, 3: 2 times, 5: 1 time

**Your Code**

```
#include <stdio.h>
int main() {
    int n, i, j, count;
        if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    int visited[n];
        for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
        visited[i] = 0;
    }
        for (i = 0; i < n; i++) {

        if (visited[i] == 1) {
            continue;
```

**Input**

```
2 2 1 5 1 3 3
```

**Output**

## CO3 - AT3 - Mock Interview

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QUESTIONS

**Q1**
CO3 - AT3 - MOCK INTERVIEW - ORAL ANSWERS
100 marks

0/1 answered

### Q1 CO3 - AT3 - MOCK INTERVIEW - ORAL ANSWERS

100 marks

1.What is the difference between an array and a variable?

2. What is meant by the base address of an array?

3.What happens if an array size is not specified during initialization?

4. How are 2D arrays stored in memory?

5. Give one real-life application of multidimensional arrays.

6.Why do arrays start from index 0 instead of 1?

7.Which engineering fields commonly use multidimensional arrays?

8.Can a multidimensional array have more than three dimensions?

9.Why are images stored as multidimensional arrays?

10. What is row-major order?

11. Does C use row-major or column-major order?

12. Why can't arrays return multiple values?

13. What happens when array bounds are exceeded?

14. Why are strings stored as character arrays?

15. Explain the difference between char str[] and char *str.

16. Can two-dimensional arrays be passed to functions?

17. When would a linked list outperform an array?

18. Differentiate gets(), fgets(), scanf(), and getchar()